Koo Foundation Sun Yat-Sen Cancer Center

A Model-Agnostic AI Agent for Automated Molecular Tumor Board Reports: A Pan-Cancer Proof-of-Concept

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Background

- Deterministic AMP/ASCO/CAP tiering[1] takes the higher tier between OncoKB[2] and CiVIC[3].
- The rule conflates knowledge-base (KB) agreement with one-sided coverage, and with high-tier CiVIC evidence in a different tumor type, risking false Tier I calls.
- LLM variant-tiering benchmarks report over-classification. We therefore restrict LLM use to KB-discordant variants only.

Evidence state	OncoKB	CiVIC	Rule output	
Both KBs agree	Level 1	Tier I A	Tier I A	✓
One KB has no assertion	Level 1	none	Tier I A	✓
Evidence from a different tumor	Level 3B	Tier I A*	Tier I A	✗

* asserted in another tumor type: a false Tier I call

Figure 1. Three evidence states receive the same call under the higher-tier rule. Only the first two warrant it.

Methods

- **Pipeline.** A `targets`-orchestrated R pipeline integrating VEP, OncoKB, CiVIC, ESCAT[4] and AMP/ASCO/CAP tiering.
- **Routing.** Concordant variants resolve deterministically. KB-discordant variants escalate to a model-agnostic agent that emits a unified tier, confidence and rationale (Table 1).
- **Benchmark.** 112 real variant–tumor cases (99 COSMIC plus 13 cross-tumor stress) against the deterministic baseline.
- **Reproducibility.** Ten cases, three runs within model and across Claude Opus 4.7, Sonnet 4.6 and Haiku 4.5 (Fleiss κ[5]).
- **Deployment.** 62 consecutive late-stage cancer patients, 8 monthly TruSight Oncology 500 batches, 11 cancer types.

Output field	Allowed values
`unified_amp_tier`	Tier I A to Tier IV (six levels)
`confidence`	high · moderate · low
`discordance_nature`	concordant, or one of five mismatch types
`primary_evidence_source`	OncoKB · CiVIC · Combined · Neither
`resolution_rationale`	two to four sentences

Table 1. JSON schema of the agent output. Every response is validated before use; the agent has no tool access.

Architecture

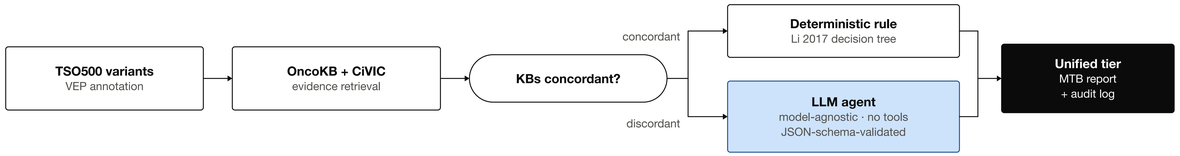


Figure 2. The LLM is invoked only where the two knowledge bases disagree. Every agent call is logged for audit.

112 / 112
Schema-valid output
JSON schema, all cases

80.4%
Tier agreement
90 of 112 versus baseline

0 / 70
Tier I/II overrides
where OncoKB is the only source

κ 1.000
Within-model
Fleiss κ, three runs

κ 0.717
Cross-model
Fleiss κ, Opus / Sonnet / Haiku

Results: agent versus rule

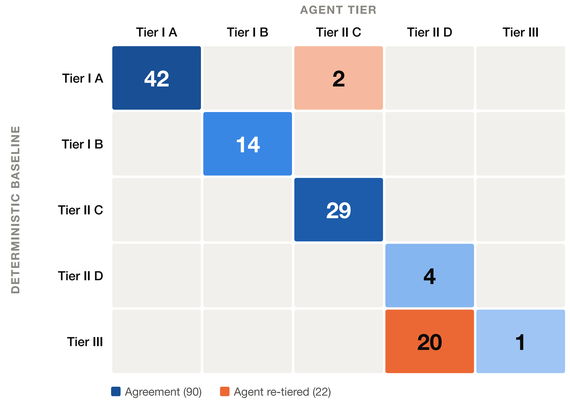


Figure 3. Deterministic baseline versus agent tier for 112 variant–tumor cases. Cells show case counts.

- Disagreements concentrated at documented rule blind spots.
- **20 of 112:** Tier III VUS to Tier II D, for oncogenic variants lacking a therapeutic level.
- **2 of 112:** Tier I A to Tier II C on tumor specificity.

Reproducibility

		Opus 4.7			Sonnet 4.6	Haiku 4.5
		run 1	run 2	run 3		
KRAS G12C	NSCLC	I A	I A	I A	I A	I A
BRAF V600E	melanoma	I A	I A	I A	I A	I A
AKT1 E17K	breast	I A	I A	I A	I A	I A
FLT3 ITD	AML	I A	I A	I A	I A	I A
RET M918T	NSCLC	II C	II C	II C	II C	II C
ALK F1174L	NSCLC	II C	II C	II C	II C	II C
TP53 R175H	breast	II D	II D	II D	II D	III
TP53 R273H	ovarian	II D	II D	II D	II D	III
APC R1450*	colon	II D	II D	II D	III	III
GATA3 R330fs	breast	II D	II D	II D	III	III

Figure 4. Tier call per case across three Opus 4.7 runs, Sonnet 4.6 and Haiku 4.5.

- Every Tier I A call and the *RET* M918T down-tier were identical across all three models.
- Cross-model variation was confined to the Tier III VUS / Tier II D boundary (Cohen κ versus Opus: Sonnet 0.722, Haiku 0.500).

Prospective deployment

- 62 consecutive late-stage patients across 11 cancer types, in 8 monthly batches.
- All 62 cases completed end-to-end, each with an auditable per-case reconciliation log.

Safety by evidence stratum

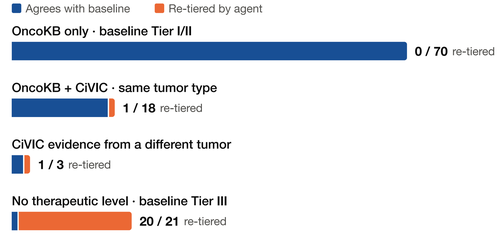


Figure 5. Re-tiering by the evidence available to the rule. No OncoKB-only Tier I/II call was overridden. The two Tier I A down-tiers were *RET* M918T in NSCLC and *BCOR* ITD in RCC (a histology call pending expert review).

Case: *RET* M918T in NSCLC

OncoKB	Level 3B: approved in a different tumor type	Tier II C
CiVIC	Tier I A, but in medullary thyroid carcinoma	Tier I A
Rule	Takes the higher tier: a false Tier I call	Tier I A
Agent	Flags a tumor-specificity mismatch and follows OncoKB	Tier II C

> "CiVIC reports TIER_I_LEVEL_A but explicitly in Medullary Thyroid Carcinoma, not NSCLC, so it cannot be applied as Tier I in the queried context."
> Agent rationale, verbatim excerpt from the audit log

Conclusions

- Confining LLM invocation to KB-discordant variants matches deterministic reproducibility on safety-critical cases.
- The agent detects tumor-specificity errors that the rule cannot encode.
- This supports deployment as second-opinion decision support in molecular tumor boards.
- Code, prompts and schemas are open source.

Limitations. Agent calls were not adjudicated by an expert panel. Reproducibility was measured on a ten-case subset. Deployment was single-center, without outcome data.

REFERENCES

1. Li MM, et al. J Mol Diagn 2017;19:4–23.
2. Chakravarty D, et al. JCO Precis Oncol 2017;1:1–16.
3. Griffith M, et al. Nat Genet 2017;49:170–174.
4. Mateo J, et al. Ann Oncol 2018;29:1895–1902.
5. Landis JR, Koch GG. Biometrics 1977;33:159–174.

DISCLOSURE

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CONTACT

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github.com/htlin222/ngs-tertiary-analysis-skills

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